UFUG 2104
Written Assignment 1

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March 9, 2026

Tips: Please clilck the link below to get to the targeted answers.

Problem 1
Problem 2
Problem 3
Problem 4
Problem 5

1 Problem 1

(16 pts) The pdf (probability density function) of X is given by

$$f_X(x) = \begin{cases} -\frac{1}{4}x, & -2 \le x < 0 \\ cx^2, & 0 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of the constant c.

(b) Plot the pdf of X from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e., where they intercept the y-axis and x-axis. (c) Find the expression for the cdf (cumulative distribution function) of X. (d) Find the expected value of X. (e) Find the probability of the event $A = \{1 \le x \le 2\}$. (f) Find the conditional pdf of X given A, $f_{X|A}(x|A)$ and plot it from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e. where they intercept the y- axis and x-axis. (g) Find the conditional expected value of X given A, $\mathbb{E}[X|A]$

Solution

(a) By the definition of pdf, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

and given the property of the function, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = \int_{-\infty}^{-2} 0dx + \int_{-2}^{0} \left(-\frac{1}{4}x\right) dx + \int_{0}^{3} (cx^2)dx + \int_{4}^{\infty} 0dx = 1$$

Which means

$$(-\frac{1}{8}x^2)\Big|_{-2}^{0} + (\frac{1}{3}cx^3)\Big|_{0}^{3} = \frac{1}{2} + 9c = 1$$

$\Rightarrow \boxed{c = \frac{1}{18}}$

(b) The pdf can be graphed as below:

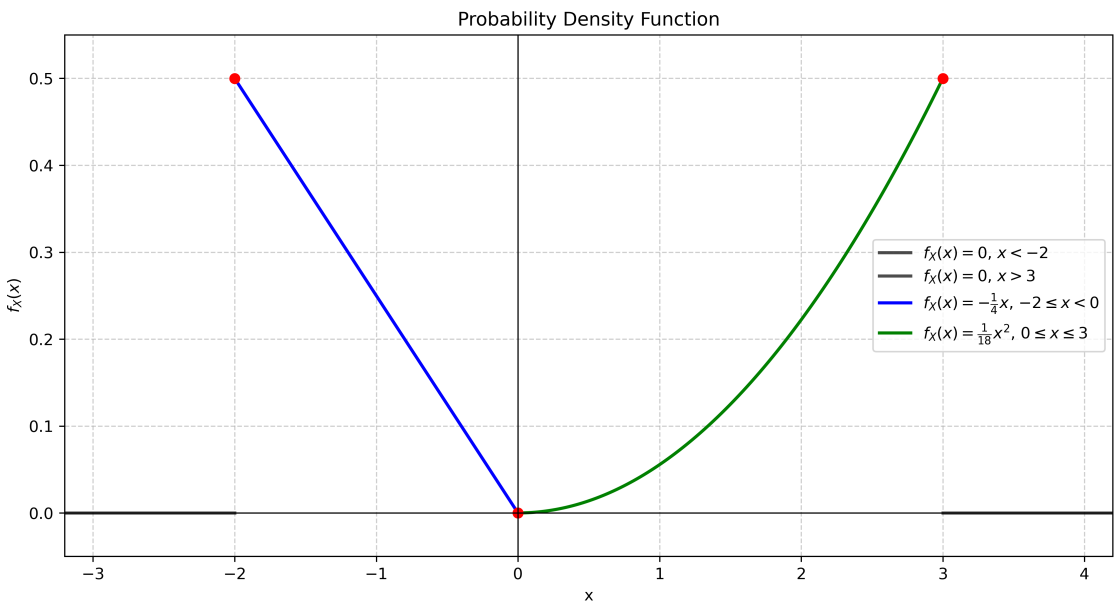

Figure 1

(c) The cdf can be calculated as below:

$$\boxed{F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{2} - \frac{1}{8}x^2 & -2 \le x < 0 \\ \frac{1}{2} + \frac{1}{54}x^3 & 0 \le x \le 3 \\ 1 & 3 < x \end{cases}}$$

(d) the expected value can be cal culated as below: when x < -2, the expected value is 0; when -2 <= x < 0, the expected value is

$$\int_{-2}^{0} -x(\frac{1}{4}x)dx = -\left(\frac{1}{12}x^3\right)\Big|_{x=-2}^{0} = -\frac{2}{3}$$

when $0 \le x \le 3$, the expectation is

$$\int_{0}^{3} x\frac{1}{18}x^2 dx = \left(\frac{1}{72}x^4\right)\Big|_{x=0}^{3} = \frac{9}{8}$$

And the same expectation value of 0m can be applied to the case when $x > 3$. So the total expectation for the expectation of

$$\boxed{\mathbb{E}(X) = -\frac{2}{3} + \frac{9}{8} = \frac{11}{24}}$$

(e) For the event $A = \{1 \le x \le 2\}$, the probability can be calculated in this way:

$$\boxed{\mathbb{P}(A) = \int_{1}^{2} \frac{1}{18}x^2 = \frac{1}{54}x^3\Big|_{x=1}^{2} = \frac{7}{54}}$$

(f) When $x \in [1, 2]$, the conditional pdf can be calculated as below:

$$f_{X|A}(x|A) = \frac{f_X(x)}{\mathbb{P}(A)} = \frac{\frac{1}{18}x^2}{\frac{7}{54}} = \frac{3}{7}x^2, \quad x \in [1, 2]$$

When $x \notin [1, 2]$, the conditional pdf can be calculated as 0. So we have:

$$f_{X|A}(x|A) = \begin{cases} \frac{3}{7}x^2 & x \in [1, 2] \\ 0 & \text{otherwise} \end{cases}$$

(g) we only have to calculate the conditional Expectation when $x \in [1, 2]$:

$$\boxed{\mathbb{E}[X|A] = \int_{1}^{2} \frac{3}{7}x^2 = \frac{9}{7}}$$

2 Problem 2

3 Problem 3

4 Problem 4

5 Problem 5